

Special or Miscellaneous Operations - Zero to Hero Questions List

May 26, 2026

1. Write a program to find the size of `int`, `float`, `char`, and `double` using `sizeof`. (*Hint: Use `sizeof(data_type)`*)
2. Write a program to find the size of an array without hardcoding the number of elements. (*Hint: Divide total array size by size of one element*)
3. Create a program that prints the memory address of a variable. (*Hint: Use address-of operator `&`*)
4. Write a program to store the address of an integer variable in a pointer and print it. (*Hint: Use `int *ptr`*)
5. Write a program to print the value of a variable using a pointer. (*Hint: Use dereference operator `*`*)
6. Swap two numbers using pointers. (*Hint: Pass addresses to a function*)
7. Write a program to access array elements using pointer arithmetic. (*Hint: `*(ptr + i)`*)
8. Find the larger number using the conditional operator `?:`. (*Hint: `(a>b)?a:b`*)
9. Find whether a number is even or odd using the ternary operator. (*Hint: `n%2==0 ?`*)
10. Write a program to find the absolute value of a number using the conditional operator. (*Hint: Check if number is negative*)
11. Write a program to check if a character is uppercase or lowercase using `?:`. (*Hint: ASCII range*)
12. Create a menu-driven calculator using the conditional operator. (*Hint: Nested ternary operators*)
13. Write a program to print the address of each array element. (*Hint: Use `&arr[i]`*)
14. Create a pointer to a pointer and print a value using double dereferencing. (*Hint: Use `**ptr`*)
15. Write a program to compare two pointer addresses. (*Hint: Use relational operators on pointers*)
16. Write a program to dynamically access string characters using pointers. (*Hint: Increment pointer*)
17. Reverse an array using pointers only. (*Hint: Use two pointers*)
18. Write a program to calculate the length of a string using pointers. (*Hint: Traverse until `'\0'`*)
19. Copy one string into another using pointers. (*Hint: Character-by-character copying*)

20. Concatenate two strings using pointers. (*Hint: Move pointer to end of first string*)
21. Create a program to demonstrate pointer increment and decrement. (*Hint: Observe address changes*)
22. Write a program to print values and addresses of variables together. (*Hint: %p for addresses*)
23. Write a program to find the largest element in an array using pointers. (*Hint: Traverse using pointer arithmetic*)
24. Use `sizeof` with structures and print total memory occupied. (*Hint: Define a structure first*)
25. Write a program to access structure members using the arrow operator `->`. (*Hint: Use structure pointer*)
26. Create a structure pointer and modify structure values through it. (*Hint: `ptr->member = value`*)
27. Write a program using the comma operator to execute multiple expressions in one statement. (*Hint: (`a=5, b=10, a+b`)*)
28. Find the sum of array elements using only pointers. (*Hint: Increment pointer in loop*)
29. Write a program to print all command-line arguments. (*Hint: Use `argv[]`*)
30. Count the number of command-line arguments entered. (*Hint: Use `argc`*)
31. Write a program to demonstrate void pointers. (*Hint: Typecast before dereferencing*)
32. Create a generic swap function using void *. (*Hint: Use typecasting and temporary storage*)
33. Write a program to print the next memory location of different data types. (*Hint: Observe pointer increments*)
34. Create a program to determine if two pointers point to the same variable. (*Hint: Compare addresses*)
35. Write a program to print a 2D array using pointers. (*Hint: Use pointer arithmetic carefully*)
36. Find the smallest element in an array using pointers. (*Hint: Traverse with pointer increment*)
37. Write a program to count vowels in a string using pointers. (*Hint: Traverse string character by character*)
38. Demonstrate the difference between `arr` and `&arr`. (*Hint: Print addresses and sizes*)
39. Write a program to allocate memory dynamically and store values. (*Hint: Use `malloc()`*)
40. Write a program to free dynamically allocated memory. (*Hint: Use `free()`*)
41. Write a program to reallocate memory using `realloc()`. (*Hint: Increase array size dynamically*)
42. Create a program to initialize memory using `calloc()`. (*Hint: Observe default values*)
43. Write a program to print values using function pointers. (*Hint: Store function address in pointer*)
44. Create a calculator using function pointers. (*Hint: Call functions through pointers*)

45. Write a program to sort an array using pointers. (*Hint: Swap using dereferencing*)
46. Write a program to demonstrate null pointers. (*Hint: Initialize with `NULL`*)
47. Create a program to check whether a pointer is null before dereferencing. (*Hint: Use `if(ptr != NULL)`*)
48. Write a program to access structure array elements using pointers. (*Hint: Use `(ptr+i)->member`*)
49. Create a program to print a string in reverse using pointers. (*Hint: Move pointer to end first*)
50. Write a program to find the difference between two pointers in an array. (*Hint: Subtract pointers*)